

Manan Arora

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Education

University of Michigan – Ann Arbor

Aug 2025 – May 2027

MS in Electrical and Computer Engineering – Specializing in Machine Learning and Signal Processing

Course Work: Natural Language Processing, Machine Learning, Reinforcement Learning, Computer Vision

Sardar Patel Inst. of Technology

Aug 2019 – Jun 2023

B.Tech. in Electronics and Telecommunication Engineering | GPA: 3.96/4.0

Professional Experience

AI Engineering Intern | Honeywell Technologies | Atlanta, GA

May 2026 – Aug 2026

- Developed a multi-agent framework using LangGraph to construct structured knowledge graphs from enterprise metadata, spanning 3 enterprise catalogs (1000+ schemas, ~10,000 tables) as the reasoning foundation for Honeywell's Edge AI platform.
- Shipped the Context Building layer for Honeywell's Cognition Edge AI platform, quantizing and exporting Qwen models via ONNX for real-time on-device inference across 30M+ assets in 5 continents.
- Won 1st place among 100+ engineers (US/EU) at the Honeywell Global Hackathon for a LangChain multi-agent framework diagnosing ERP-linked failures and alerting field engineers.

AI Engineer | AuxoAI | San Francisco, CA

Aug 2023 – Jun 2025

- Built retrieval and LLM tool-execution capabilities for Damia, AuxoAI's enterprise AI platform, enabling AI analysis over 10,000+ page document corpora with 96% accuracy, serving 3M MAU.
- Shipped a Natural Language to SQL platform serving 1M+ users across 45 enterprises, by building distributed Lambda functions to improve time to output to under 10 seconds.
- Collaborated with product managers, domain experts, and customers to translate business requirements into production AI solutions across enterprise manufacturing workflows.

Research and Publications

Researcher - Jason Corso Lab | University of Michigan | Ann Arbor, MI

Jan 2026 – Present

- Fine-tuning Qwen and InternVL multimodal foundation models at test time using Group Relative Policy Optimization (GRPO)-based reinforcement learning, with frequency/diversity reward signals, achieving 80% increase in model accuracy.
- Developed multimodal evaluation infrastructure with 30+ scripts and multi-GPU sharding, benchmarking YOLO-family detectors and VLMs on MS-COCO to evaluate multimodal reasoning and inference-time intervention effectiveness.

Research Assistant - Ziyu Song Lab | University of Michigan | Ann Arbor, MI

Oct 2025 – Jan 2026

- Researched ML-based remaining useful life (RUL) prediction for second-life EV batteries, modeling degradation mechanisms from electrochemical impedance spectroscopy (EIS) data through physics-informed feature engineering.
- Benchmarked 6 architectures (ANN, CNN, XGBoost, CatBoost, contrastive deep learning ensemble) with Optuna hyperparameter tuning, achieving 3.69% RMSE on held-out data, with the best ensemble outperforming individual baselines across all degradation regimes.

Reinforcement Learning: Adaptive MAPPO | Python, PyTorch, TorchRL

[Paper](#) · [Code](#)

- Developed a multi-agent RL algorithm that dynamically balances centralized and decentralized value estimates, evaluating it on 72 experiments and achieving a 21% late-training drop vs. 71% for the baseline.

Skills and Technologies

Languages & Infra: Python, Java, C++, JavaScript, Typescript, React, SQL, Matlab, NumPy, pandas, Google Cloud Platform, Microsoft Azure, Docker, Kubernetes, Linux, PostgreSQL, MongoDB, Simulink

ML/AI: PyTorch, TensorFlow, TorchRL, Torchvision, Reinforcement Learning, Hugging Face, vLLM, OpenAI SDK, LangGraph, RAG, Vision-Language Models, Inference-Time Optimization, Knowledge Graphs (RDF/OWL), OpenCV